Quantum Noise Sensitivity Mapping Engine: Density-Matrix Sensitivity Analysis Under Local Amplitude and Phase Damping

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Abstract

Quantum information processing depends on the preparation and preservation of coherent quantum states, yet realistic systems are inevitably subject to environmental noise. This paper presents a dissertation-style analysis of the *Quantum Noise Sensitivity Mapping Engine*, a compact Python simulator designed to study state degradation using density matrices and explicitly implemented Kraus channels. The engine models a single-qubit superposition state, a Bell state, and GHZ states with two to four qubits, and it applies local amplitude damping and phase damping noise without relying on external quantum-computing frameworks. The simulation pipeline constructs ideal density matrices, embeds single-qubit Kraus operators into multipartite Hilbert spaces, applies noise independently to each qubit, validates the resulting density matrices, and computes fidelity and purity. Parameter sweeps over noise strength and a GHZ scaling study over qubit number are exported as CSV files and SVG figures. The numerical outputs show that fidelity decreases as noise increases in all bundled studies, while purity distinguishes the physical action of the two channels: phase damping drives states toward mixedness, whereas strong amplitude damping can restore purity by relaxing the system toward a pure ground state even as fidelity to the target state continues to decline. The GHZ study additionally shows a consistent decrease in robustness with increasing qubit number under phase damping. The project demonstrates that a small, transparent, framework-free implementation can still support a technically meaningful study of open-system quantum behavior while remaining readable, extensible, and pedagogically valuable.

1 Introduction

Quantum computing derives its potential from nonclassical state structure, particularly superposition and entanglement. These features allow the state space of a system to grow rapidly with qubit number and enable operational effects that do not have classical analogues. Their usefulness, however, depends on maintaining coherence. Real quantum systems are open systems: they interact with surrounding degrees of freedom, and these interactions degrade the quantum properties that computation and communication protocols require [1]. The practical challenge is therefore not only to prepare quantum states, but also to understand how quickly and in what manner they lose their defining features under noise.

Noise analysis can be approached at different levels of abstraction. Large software stacks and hardware-oriented toolchains are essential for realistic workflows, but they can also obscure the mathematical structure of the model being used. For many research and educational purposes, there is value in a smaller simulator whose assumptions are explicit and whose implementation can be traced directly back to the underlying equations. The *Quantum Noise Sensitivity Mapping Engine* was built with this goal in mind. It is intentionally small, supports only one to four qubits, avoids external quantum frameworks, and performs state evolution entirely through density matrices and manually defined Kraus operators.

This design choice is not merely stylistic. Density matrices are the correct representation for irreversible noise because open-system channels can transform pure states into mixed states. A pure-state or statevector-only picture is insufficient for such dynamics. Kraus operators provide a mathematically standard and computationally direct method for implementing trace-preserving quantum channels [2]. When these ideas are expressed in a transparent codebase, the simulator becomes more than a numerical utility: it becomes a bridge between open-system quantum theory and reproducible computation.

The present work analyzes this engine as both a scientific model and a software artifact. The aim is to determine how different representative quantum states degrade under amplitude damping and phase damping, and how that degradation depends on system size. The states considered are a single-qubit superposition state, a Bell state, and GHZ states with two to four qubits. The channels are applied independently to each qubit, and the resulting states are evaluated using fidelity with respect to the ideal target state and the purity of the noisy density matrix.

The scope is deliberately limited. The engine does not attempt gate-level circuit simulation, does not include hardware calibration, and does not benchmark against full-featured frameworks. Instead, it emphasizes mathematical correctness, interpretability, and extensibility. This paper therefore treats the project as a small but rigorous numerical study: one whose value lies not in large-scale simulation, but in its clarity, traceability, and ability to expose the structure of quantum-state degradation under simple local noise models.

2 Project Scope and Design Intent

Before entering the formal theory and implementation details, it is useful to state clearly what the engine is and what it is not. The README and source code show that the project is a *framework-free quantum noise simulator* whose principal purpose is to map the sensitivity of a small set of states to two canonical local decoherence channels. All internal state representations are density matrices, all noise is implemented by hand through Kraus operators, and the experiment runner is restricted to a compact suite of bundled studies.

This design reflects four explicit intentions.

1. **Use density matrices throughout.** The code does not include a statevector pathway, reinforcing the fact that the project is aimed at mixed-state and decoherence analysis rather than unitary-only evolution.

2. **Implement channels manually.** The noise model is not delegated to a library call. The simulator constructs Kraus operators directly and embeds them into the full Hilbert space through tensor products.

3. **Keep the system small.** The upper bound of four qubits is chosen to avoid the loss of readability that accompanies exponentially growing density matrices.

4. **Prioritize transparent data products.** Each bundled experiment writes human-readable CSV data and SVG figures, allowing the user to inspect both the numerical arrays and the rendered trends.

In methodological terms, the one-to-four-qubit limit should not be interpreted as a weakness of the project. A density matrix for n qubits has dimension $2^n \times 2^n$, which means the number of matrix elements grows as 4^n. Within that scaling, even four qubits are sufficient to exhibit genuinely multipartite behavior while still remaining small enough for explicit operator construction, manual inspection, and algebraic verification. The engine therefore occupies a useful middle ground: it is more physically faithful than a pure-state toy model, yet more transparent than a large black-box framework.

3 Theoretical Foundations

3.1 Pure states, mixed states, and the need for density matrices

A pure quantum state $|\psi\rangle$ may be represented either as a statevector or as the rank-one density matrix

$$\rho = |\psi\rangle\langle\psi|. \tag{1}$$

This representation becomes indispensable in open-system settings because noise processes generally produce statistical mixtures rather than single pure states. A mixed state is written as

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \qquad p_i \geq 0, \qquad \sum_i p_i = 1. \tag{2}$$

Unlike a statevector, the density matrix can describe incoherent classical uncertainty and decoherence-induced loss of phase information in the same formal object.

The physics of decoherence makes this distinction necessary. Environmental interaction suppresses the off-diagonal structure that encodes coherence and can drive a system away from an intended target state, even in the absence of any measurement by the user [1]. Since the present engine is designed specifically to model such irreversible degradation, a density-matrix-only representation is not merely compatible with the goal of the project; it is the appropriate mathematical framework.

3.2 Physicality conditions for density matrices

For a matrix to represent a physical quantum state, three conditions must hold:

$$\rho = \rho^\dagger, \tag{3}$$

$$\mathrm{Tr}(\rho) = 1, \tag{4}$$

$$\rho \succeq 0, \tag{5}$$

where Eq. (5) denotes positive semidefiniteness. These conditions have direct computational significance in the project. The metrics module checks Hermiticity numerically, checks that the trace is approximately unity, and computes eigenvalues to ensure that no negative spectrum appears beyond a numerical tolerance. Thus, the code does not merely compute on matrices that are *assumed* to be physical; it actively verifies that they remain physically admissible after channel application.

3.3 Tensor products and multipartite state construction

Multipartite states are built through tensor products. For computational-basis states,

$$|b_1 b_2 \cdots b_n\rangle = |b_1\rangle \otimes |b_2\rangle \otimes \cdots \otimes |b_n\rangle, \qquad b_i \in \{0, 1\}. \tag{6}$$

The state-construction module implements this using repeated Kronecker products. This approach is used both for basis-state construction and for the assembly of the supported entangled states. The engine studies:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \tag{7}$$

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \tag{8}$$

$$|\mathrm{GHZ}_n\rangle = \frac{|0\cdots0\rangle + |1\cdots1\rangle}{\sqrt{2}}, \qquad 2 \leq n \leq 4. \tag{9}$$

Each of these is converted immediately into the density-matrix form of Eq. (1).

3.4 Kraus operators and channel action

The simulator models noise through the operator-sum representation

$$\rho' = \sum_k E_k \rho E_k^\dagger, \tag{10}$$

subject to the completeness relation

$$\sum_k E_k^\dagger E_k = I. \tag{11}$$

This representation describes completely positive, trace-preserving evolution and is standard in open-system quantum information theory [2]. Within the present project, it has an additional advantage: it maps naturally onto explicit linear algebra operations and therefore onto short, inspectable Python code.

3.5 Amplitude damping

Amplitude damping models relaxation from $|1\rangle$ to $|0\rangle$, which may be interpreted as energy loss to an environment. The implementation uses the Kraus set

$$E_0^{(A)} = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \qquad E_1^{(A)} = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}, \qquad 0 \leq \gamma \leq 1. \tag{12}$$

The parameter γ is therefore a local damping strength. Because this channel changes populations as well as coherences, it can move a state toward the ground state while simultaneously reducing its overlap with the initial target.

3.6 Phase damping

Phase damping, also called dephasing, suppresses coherence without changing the computational-basis populations. The project uses the three-operator representation

$$E_0^{(P)} = \sqrt{1-\lambda}\, I, \qquad E_1^{(P)} = \sqrt{\lambda}\, |0\rangle\langle 0|, \qquad E_2^{(P)} = \sqrt{\lambda}\, |1\rangle\langle 1|, \qquad 0 \leq \lambda \leq 1. \tag{13}$$

This channel is especially relevant to GHZ-type states because their nonclassical character depends directly on the off-diagonal coherence between $|0 \cdots 0\rangle$ and $|1 \cdots 1\rangle$.

3.7 Fidelity and purity

The engine computes fidelity against a pure reference state and the purity of the evolved density matrix. Since the reference state ρ_{ideal} is always pure in the bundled experiments, fidelity is evaluated as

$$F(\rho, \rho_{\text{ideal}}) = \mathrm{Tr}(\rho \rho_{\text{ideal}}). \tag{14}$$

Purity is given by

$$P(\rho) = \mathrm{Tr}(\rho^2). \tag{15}$$

These two metrics answer different questions. Fidelity asks whether the state remains close to the intended target, whereas purity asks whether the state remains nearly pure. The results of this project show explicitly that these questions can yield different conclusions for the same noisy evolution.

4 Implementation Architecture

4.1 Module-by-module structure

The project is architected to follow the mathematical workflow closely. Table 1 summarizes the role of each source file.

4.2 Data flow through the engine

The internal data flow can be described as a sequence of mathematically interpretable steps:

1. construct an ideal state as a density matrix;
2. define the local Kraus operators for the selected channel;
3. expand each local operator into the full Hilbert space;
4. apply the channel sequentially to each qubit;
5. numerically stabilize the result and check physicality;
6. compute fidelity and purity;

Table 1: Responsibilities of the main source modules.

Module	Main responsibility
`core/states.py`	Builds basis vectors, tensor products, and the density matrices for the single-qubit superposition, Bell, and GHZ states.
`core/noise.py`	Defines single-qubit Kraus operators, validates channel parameters, embeds local operators into multipartite Hilbert spaces, applies channels, and numerically stabilizes the output density matrix.
`core/metrics.py`	Validates physicality and computes fidelity and purity.
`core/evolve.py`	Selects the requested noise model and applies the channel sequentially to all qubits in the state.
`experiments/sweep.py`	Performs one-dimensional parameter sweeps over noise strength and scaling sweeps over qubit number.
`experiments/run_experiments.py`	Defines the bundled experiments, writes CSV outputs, manages the output directories, and saves plot files.
`visualization/plots.py`	Generates fidelity curves, purity curves, and the GHZ heatmap using `matplotlib`.
`main.py`	Executes the full experiment suite and prints the saved file paths.

7. repeat across parameter grids;
8. export arrays and figures.

This flow is reflected directly in the directory structure. The design avoids unnecessary abstraction layers, which is a significant methodological advantage for a project that is meant to be read, verified, and extended by users who want to understand the mechanics of noisy density-matrix evolution rather than merely invoke it.

4.3 Framework-free implementation and its significance

The project deliberately avoids Qiskit, QuTiP, and other quantum libraries. This matters for two reasons. First, the resulting code makes the mathematical content explicit. The amplitude-damping matrices, phase-damping matrices, tensor products, and trace operations appear plainly in the source, which makes validation and review easier. Second, the absence of an external framework forces the project to confront numerical details directly, such as trace renormalization, Hermiticity restoration, and shape checks. In a larger framework these details are often handled internally; here they are visible and therefore open to scrutiny.

4.4 The four-qubit limit as a methodological choice

The upper limit of four qubits is often easy to misread as a lack of ambition. In reality, it is part of the project methodology. Because a density matrix has 4^n entries, even moderate increases in qubit number rapidly obscure the exact operator structure of the simulation. The four-qubit limit ensures that the engine remains computationally manageable and, more importantly, analytically inspectable. This allows the bundled studies to be checked directly against closed-form reasoning, which is one of the strongest features of the present implementation.

5 Methodology

5.1 Simulation workflow

The complete simulation procedure implemented by the project is as follows:

1. Construct the ideal state vector in the computational basis.

2. Convert the ideal state to a density matrix.
3. Build the single-qubit Kraus operators for the chosen noise channel.
4. Embed each single-qubit Kraus operator into the full n-qubit Hilbert space using tensor products.
5. Apply the local channel independently and sequentially to each qubit.
6. Symmetrize the output matrix and renormalize its trace.
7. Validate Hermiticity, unit trace, and positive semidefiniteness numerically.
8. Compute fidelity and purity.
9. Sweep the noise parameter or the qubit count according to the experiment definition.
10. Save the numerical outputs as CSV files and generate SVG plot files.

This workflow is not merely conceptual; it corresponds directly to the actual control flow of `states.py`, `noise.py`, `evolve.py`, `metrics.py`, `sweep.py`, and `run_experiments.py`.

5.2 Exact states used in the bundled experiments

The experiments are built around three concrete state families:

- the single-qubit $|+\rangle$ state of Eq. (7),
- the Bell state $|\Phi^+\rangle$ of Eq. (8),
- GHZ states $|\mathrm{GHZ}_n\rangle$ for $n = 2, 3, 4$ from Eq. (9).

No additional states are included in the current engine. The GHZ constructor itself enforces the range $2 \leq n \leq 4$, so the upper bound is an actual code-level restriction and not simply a statement in the documentation.

5.3 Multi-qubit channel application

For an n-qubit density matrix ρ, the code first computes the expected dimension $2^n \times 2^n$ and verifies that the input matrix has the correct shape. For each qubit index j, every single-qubit Kraus operator E_k is lifted to

$$\widetilde{E}_{k,j} = I \otimes \cdots \otimes I \otimes E_k \otimes I \otimes \cdots \otimes I, \tag{16}$$

and the update

$$\rho \mapsto \sum_k \widetilde{E}_{k,j} \rho \widetilde{E}_{k,j}^{\dagger} \tag{17}$$

is performed. The evolved state is then passed to the next qubit. In other words, the implementation realizes independent local noise by composing local channels sequentially across the qubit register.

5.4 Bundled experiment suite

The project defines three experiments. Their numerical configuration is fixed in `run_experiments.py` and summarized in Table 2.

The single-qubit experiment isolates the effect of amplitude damping on a minimal coherent superposition. The Bell comparison holds the initial state fixed and changes the channel type. The GHZ study changes system size while holding the channel structure fixed, making it the engine's direct scaling experiment.

Table 2: Numerical configuration of the bundled experiments.

Experiment	State(s)	Channel(s)	Parameter specification
Single-qubit amplitude study	$\|+\rangle$	Amplitude damping	$\gamma = 0.00, 0.02, \ldots, 1.00$ (51 values)
Bell-state comparison	$\|\Phi^+\rangle$	Amplitude damping and phase damping	$\gamma, \lambda = 0.00, 0.02, \ldots, 1.00$ (51 values each)
GHZ scaling study	$\|\mathrm{GHZ}_n\rangle$, $n = 2, 3, 4$	Phase damping	fixed $\lambda = 0.35$ for scaling; $\lambda = 0.00, 0.02, \ldots, 1.00$ for heatmap

5.5 Output generation

Each experiment writes numerical results using `numpy.savetxt`. The line-sweep CSV files contain columns for the noise parameter, fidelity, and purity. The GHZ heatmap CSV stores the noise parameters in the first row and the qubit counts in the first column, with the fidelity matrix occupying the interior. Plot generation is delegated to `visualization/plots.py`. Fidelity and purity sweeps are plotted as line graphs, and the GHZ heatmap is rendered via `pcolormesh`, with the heatmap axes computed from cell-edge coordinates rather than raw sample points.

6 Analytical Cross-Checks Derived from the Implemented Model

One of the advantages of the small system size is that several of the bundled results can be derived in closed form directly from the implemented channels. These derivations are not additional experiments; they are analytical checks of the same model realized by the code.

6.1 Single-qubit amplitude damping

Applying Eq. (12) to the initial density matrix $\rho_+ = |+\rangle\langle+|$ gives

$$\rho_+^{(A)}(\gamma) = \frac{1}{2}\begin{pmatrix} 1+\gamma & \sqrt{1-\gamma} \\ \sqrt{1-\gamma} & 1-\gamma \end{pmatrix}. \tag{18}$$

From Eqs. (14) and (15), the fidelity and purity are

$$F_+^{(A)}(\gamma) = \frac{1+\sqrt{1-\gamma}}{2}, \tag{19}$$

$$P_+^{(A)}(\gamma) = 1 - \frac{\gamma}{2} + \frac{\gamma^2}{2}. \tag{20}$$

Equation (20) explains the curvature observed in the saved CSV output: the purity reaches its minimum at $\gamma = 0.5$ and then increases again as the state relaxes toward $|0\rangle\langle 0|$.

6.2 Bell state under amplitude damping

For the Bell state $|\Phi^+\rangle$, the local amplitude-damping channel on both qubits produces the density matrix

$$\rho_{\Phi^+}^{(A)}(\gamma) = \frac{1}{2}\begin{pmatrix} 1+\gamma^2 & 0 & 0 & 1-\gamma \\ 0 & \gamma(1-\gamma) & 0 & 0 \\ 0 & 0 & \gamma(1-\gamma) & 0 \\ 1-\gamma & 0 & 0 & (1-\gamma)^2 \end{pmatrix}. \tag{21}$$

The resulting fidelity and purity are

$$F_{\Phi^+}^{(A)}(\gamma) = 1 - \gamma + \frac{\gamma^2}{2}, \tag{22}$$

$$P_{\Phi^+}^{(A)}(\gamma) = 1 - 2\gamma + 3\gamma^2 - 2\gamma^3 + \gamma^4. \tag{23}$$

These expressions match the symmetric purity curve observed in the amplitude-damping CSV output.

6.3 Bell and GHZ states under phase damping

Because the Bell state is a two-qubit GHZ state, local phase damping yields a simple general pattern. For the GHZ family,

$$\rho^{(P)}_{\mathrm{GHZ}_n}(\lambda) = \frac{1}{2}|0\cdots 0\rangle\langle 0\cdots 0| + \frac{1}{2}|1\cdots 1\rangle\langle 1\cdots 1| + \frac{(1-\lambda)^n}{2}|0\cdots 0\rangle\langle 1\cdots 1| + \frac{(1-\lambda)^n}{2}|1\cdots 1\rangle\langle 0\cdots 0|. \quad (24)$$

The off-diagonal term acquires a factor $(1-\lambda)^n$ because each local dephasing step multiplies coherence by $(1-\lambda)$. The corresponding fidelity and purity are

$$F^{(P)}_{\mathrm{GHZ}_n}(\lambda) = \frac{1+(1-\lambda)^n}{2}, \quad (25)$$

$$P^{(P)}_{\mathrm{GHZ}_n}(\lambda) = \frac{1+(1-\lambda)^{2n}}{2}. \quad (26)$$

For $n = 2$, Eqs. (25) and (26) describe the Bell-state phase-damping output exactly. These closed forms also reproduce the entire GHZ heatmap and the fixed-λ scaling table.

7 Results

7.1 Single-Qubit Amplitude Damping

The single-qubit experiment sweeps the amplitude-damping strength across 51 equally spaced points from 0 to 1. Figure 1 shows the fidelity relative to the initial $|+\rangle$ state, and Figure 2 shows the purity of the evolved density matrix.

The saved data in the single-qubit amplitude sweep CSV file show that fidelity decreases monotonically throughout the sweep. The fidelity is approximately 1.0000 at $\gamma = 0$, 0.99497 at $\gamma = 0.02$, 0.94721 at $\gamma = 0.20$, 0.85355 at $\gamma = 0.50$, and 0.5000 at $\gamma = 1.00$. The purity follows a different trajectory. It begins at 1.0000, falls to 0.9200 at $\gamma = 0.20$, reaches 0.8750 at $\gamma = 0.50$, and then rises back to 1.0000 at $\gamma = 1.00$. Thus, the numerical output exhibits monotonic fidelity loss but non-monotonic purity.

These values are consistent with the exact expressions in Eqs. (19) and (20). The CSV output is therefore not only internally consistent as data, but also analytically consistent with the channel model actually implemented in the code.

7.2 Bell State Noise Comparison

The Bell-state experiment compares amplitude damping and phase damping using the same parameter grid. Figure 3 shows the fidelity curves, while Figure 4 shows the purity curves.

The most striking feature of this dataset is that the Bell-state fidelity curves under amplitude damping and phase damping are numerically identical up to floating-point precision across the full sweep. A direct comparison of the two CSV files gives a maximum absolute fidelity difference of approximately 2.22×10^{-16}, which is a numerical zero at double precision. At the midpoint of the sweep, both channels yield a fidelity of 0.6250, and at maximal noise both give 0.5000.

The purity curves do *not* coincide. Under amplitude damping, the Bell-state purity begins at 1.0000, decreases to 0.5625 at $\gamma = 0.50$, and returns to 1.0000 at $\gamma = 1.00$. Under phase damping, the purity begins at 1.0000, decreases to 0.53125 at $\lambda = 0.50$, and reaches 0.5000 at $\lambda = 1.00$. The maximum absolute difference between the two purity curves is 0.5, attained at maximal noise. Thus, the experiment reveals a strong channel dependence in mixedness despite identical target-state fidelity values.

7.3 GHZ Scaling Study

The GHZ scaling study fixes the phase-damping parameter at $\lambda = 0.35$ and evaluates GHZ states of size $n =$ 2, 3, and 4. The resulting fidelity and purity values are written to `outputs/data/ghz_phase_scaling.csv` and visualized in Figure 5.

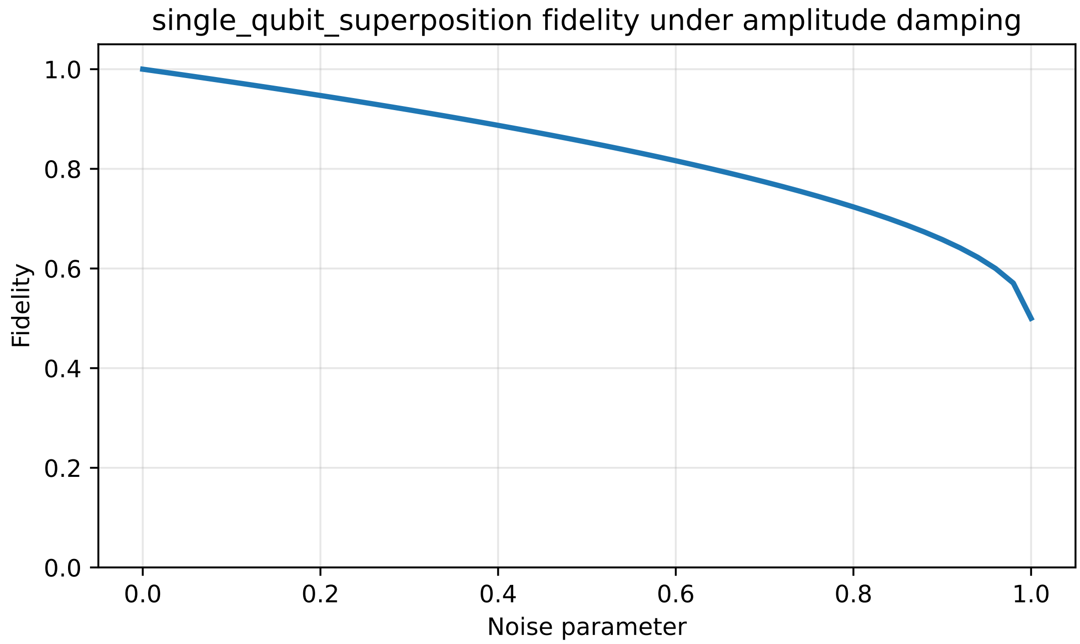

Figure 1: Fidelity of the single-qubit superposition state under amplitude damping. The plotted values are generated from `outputs/data/single_qubit_amplitude_sweep.csv`.

Table 3: GHZ scaling results under phase damping at fixed $\lambda = 0.35$.

Number of qubits n	Fidelity	Purity
2	0.71125	0.589253125
3	0.6373125	0.5377094453125
4	0.589253125	0.5159322406445312

The saved numerical values show a systematic decline in robustness as the number of qubits increases. Fidelity decreases from 0.71125 at $n = 2$ to 0.6373125 at $n = 3$ and 0.589253125 at $n = 4$. Purity decreases in parallel, from 0.589253125 to 0.5377094453125 and then to 0.5159322406445312. These values agree exactly with Eqs. (25) and (26) evaluated at $\lambda = 0.35$.

The broader heatmap, stored in `outputs/data/ghz_phase_fidelity_heatmap.csv` and shown in Figure 6, extends the same trend across the entire parameter range. At every sampled nonzero phase-noise value, larger GHZ systems have lower fidelity than smaller ones. For instance, at $\lambda = 0.20$ the fidelities are 0.8200, 0.7560, and 0.7048 for $n = 2$, 3, and 4, respectively. At $\lambda = 0.50$, they are 0.6250, 0.5625, and 0.53125. At $\lambda = 0.80$, they are 0.5200, 0.5040, and 0.5008. The heatmap therefore shows both parameter sensitivity and size sensitivity within the same visualization.

8 Discussion

8.1 Why fidelity degrades under both channels

Fidelity declines under both amplitude damping and phase damping because both channels reduce the overlap between the evolved state and the initial target state. The two channels do this differently. Amplitude damping changes populations and coherence simultaneously by driving probability mass toward the ground state. Phase damping leaves basis populations unchanged but suppresses off-diagonal structure. In either case, the evolved density matrix departs from the ideal reference, and Eq. (14) decreases accordingly. The data confirm this for all bundled studies.

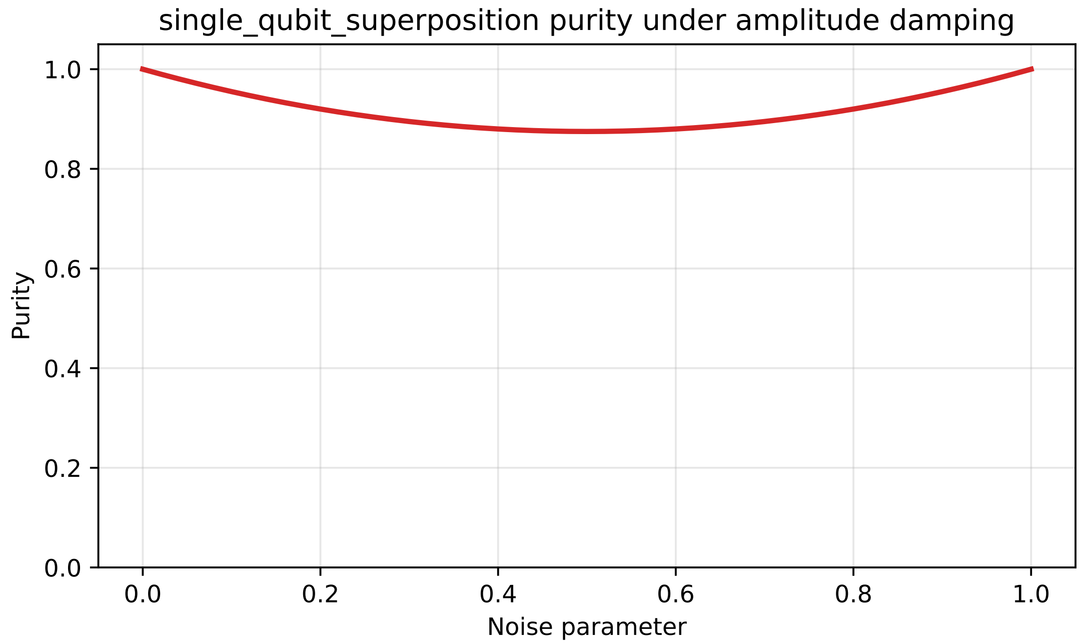

Figure 2: Purity of the single-qubit superposition state under amplitude damping. The curve decreases initially and then rises back to 1 at maximal damping.

8.2 Why purity behaves differently under the two channels

The purity results show that channel identity matters at least as much as noise magnitude. Under phase damping, purity declines because the channel suppresses coherence without relaxing the state toward a single pure computational-basis outcome. The Bell-state phase-damping curve and the GHZ scaling data both exhibit this behavior. Under amplitude damping, however, the final state can approach the pure ground state $|0\rangle\langle 0|$ or $|00\rangle\langle 00|$ even though that final state differs substantially from the intended target. This is why the single-qubit and Bell-state amplitude-damping curves return to purity 1 at maximal noise.

This distinction is scientifically important. A state may be pure and yet operationally wrong for the task for which it was prepared. In the present project, strong amplitude damping produces exactly this situation. The final state is pure, but fidelity to the original target state has fallen to 0.5. The simulator therefore illustrates a useful conceptual lesson: purity is not a proxy for correctness, and fidelity is not a proxy for mixedness.

8.3 Entanglement fragility in the Bell and GHZ studies

The Bell and GHZ experiments show how sensitivity increases when the relevant coherence is distributed across multiple qubits. The Bell state already requires phase coherence between $|00\rangle$ and $|11\rangle$. The GHZ family extends this same coherence pattern to larger systems. Under local noise, each additional qubit becomes another site at which coherence can be degraded. This is consistent with the broader literature on entanglement fragility and multiqubit robustness under decoherence [3, 4].

The GHZ heatmap makes this scaling effect visually explicit. For phase damping, the coherence factor is multiplied by $(1-\lambda)^n$, so sensitivity increases with qubit number even when the local channel strength is held fixed. The engine therefore provides a concrete numerical demonstration of a general physical principle: global coherence becomes more difficult to preserve as multipartite structure grows.

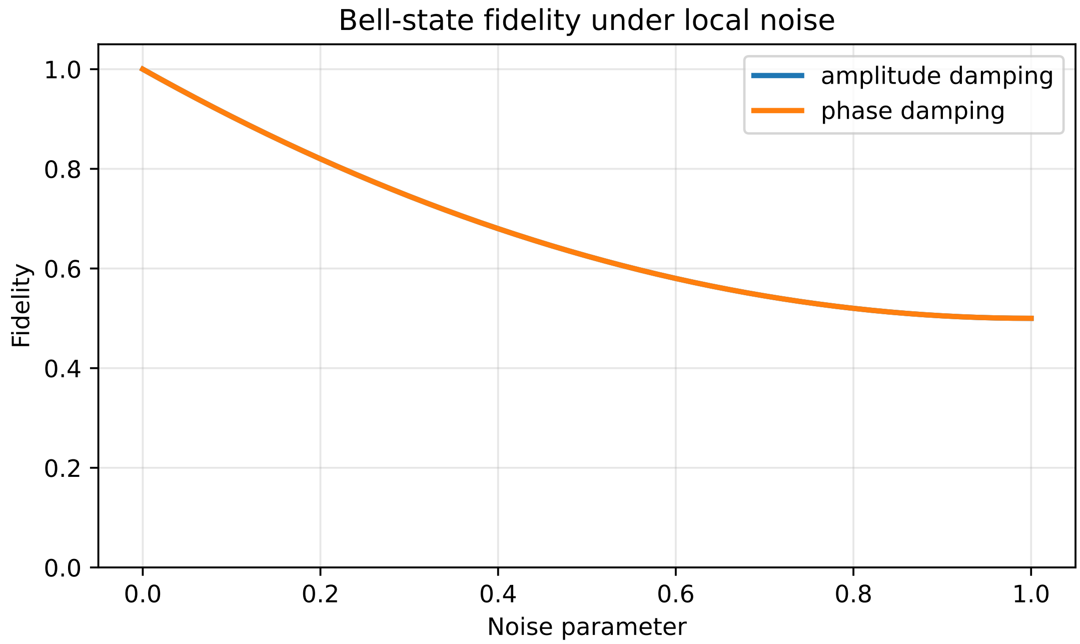

Figure 3: Bell-state fidelity under amplitude damping and phase damping. In the saved data, the two curves are numerically identical over the sampled parameter range.

8.4 What is physically meaningful and what is simulator-dependent

Some conclusions of this study are physically meaningful beyond the code itself. The distinction between dephasing and relaxation, the difference between fidelity and purity, and the scaling vulnerability of GHZ coherence all reflect genuine features of open quantum systems. Other conclusions are tied to the simplifications of the simulator. The independent local-channel assumption, the absence of gate-level dynamics, and the lack of correlated or non-Markovian noise mean that the results should not be interpreted as direct hardware predictions. Instead, they should be interpreted as clean channel-response studies under controlled assumptions.

This distinction is important for scientific honesty. The engine is strong precisely because it is explicit about what it models. It does not claim to be a full-device simulator. What it does provide is a reliable small-scale environment in which the effects of two important local channels can be studied in detail.

9 Limitations

The project has several limitations that shape the interpretation of the results:

1. The simulation is restricted to one to four qubits, which prevents direct study of larger multipartite systems.
2. The engine uses density matrices exclusively and does not provide a statevector pathway for isolated-system comparison.
3. Only two channels are implemented: amplitude damping and phase damping.
4. The channels are local and independent; correlated noise is not included.
5. No non-Markovian memory effects are modeled.
6. The project is not a gate-level circuit simulator and does not represent algorithmic execution.

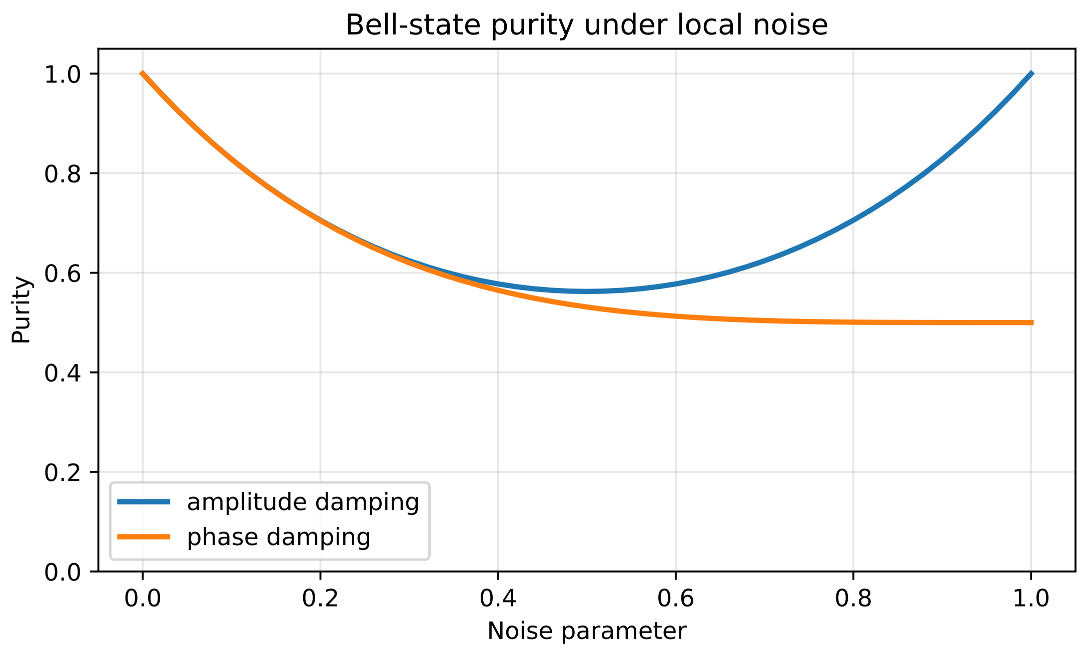

Figure 4: Bell-state purity under amplitude damping and phase damping. Unlike fidelity, purity separates the relaxation channel from the dephasing channel clearly.

7. No hardware calibration data are incorporated.
8. The experiment suite is intentionally small and pedagogical, prioritizing clarity over breadth.
9. No direct comparison against external frameworks is included in the present implementation.

These limitations do not invalidate the engine. Rather, they define its intended domain: small, transparent, physically interpretable studies of local noise in density-matrix form.

10 Future Work

The code structure suggests several technically natural extensions.

1. **Additional channels.** Depolarizing, bit-flip, phase-flip, and generalized amplitude-damping models could be added to the noise module without changing the rest of the architecture.
2. **Correlated and non-Markovian noise.** Extending the current local-channel model to correlated environments or memory-bearing channels would make the simulator more realistic and would test which of the present trends are robust.
3. **Additional state families.** W states, cluster states, and partially entangled states would broaden the notion of robustness beyond Bell and GHZ structure.
4. **Additional metrics.** Entanglement-specific measures, trace distance, relative entropy, or quantum Fisher information could supplement fidelity and purity.
5. **Repeated schedules.** Applying channels iteratively over multiple rounds would allow the study of cumulative degradation rather than single-pass parameter sweeps.
6. **Framework comparison.** Benchmarking equivalent cases against Qiskit or QuTiP would test numerical agreement while clarifying the cost of higher-level abstraction.

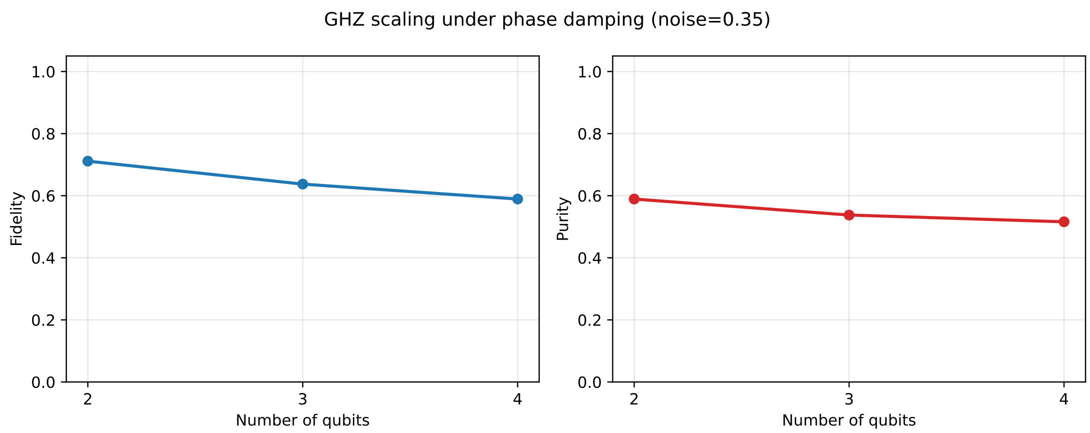

Figure 5: Fidelity and purity of GHZ states under phase damping at fixed $\lambda = 0.35$. Both quantities decrease as qubit number increases from two to four.

7. **Larger systems.** Sparse methods, tensor-network approximations, or structured state ansätze could extend the present ideas beyond dense four-qubit density matrices.

11 Conclusion

The Quantum Noise Sensitivity Mapping Engine is a compact but technically coherent simulator for studying how quantum states degrade under local noise. Its main accomplishment is not scale, but clarity. By representing states as density matrices, implementing Kraus channels manually, validating physicality after evolution, and exporting both raw data and figures, the engine creates a direct connection between quantum-noise theory and reproducible computation.

The experiments show three central findings. First, fidelity decreases with increasing noise in all bundled studies. Second, purity distinguishes amplitude damping from phase damping in a physically meaningful way: dephasing drives states toward mixedness, while strong amplitude damping can restore purity by relaxing the system toward the ground state even as fidelity to the original target continues to fall. Third, GHZ-state robustness decreases with qubit number under phase damping, demonstrating the scaling challenge associated with multipartite coherence.

The implementation is educationally valuable because it makes the mathematics visible in the code, and it is scientifically useful because it produces interpretable sensitivity maps for canonical states under canonical channels. For this class of problem, density-matrix-based simulation is not only sufficient; it is the correct lens through which irreversible quantum noise should be studied.

A Reproducibility Notes

A.1 Software dependencies

The simulator depends only on `numpy` and `matplotlib`. The main experiment entry point is `main.py`, which calls `run_all_experiments()` from `project/experiments/run_experiments.py`. The paper figures used here are generated separately in the `paper/` directory from the same underlying numerical model.

A.2 Deterministic execution

All bundled experiments are deterministic. There is no stochastic sampling, no random seed, and no Monte Carlo component in the current engine. Re-running the project reproduces the same CSV values up to ordinary floating-point precision.

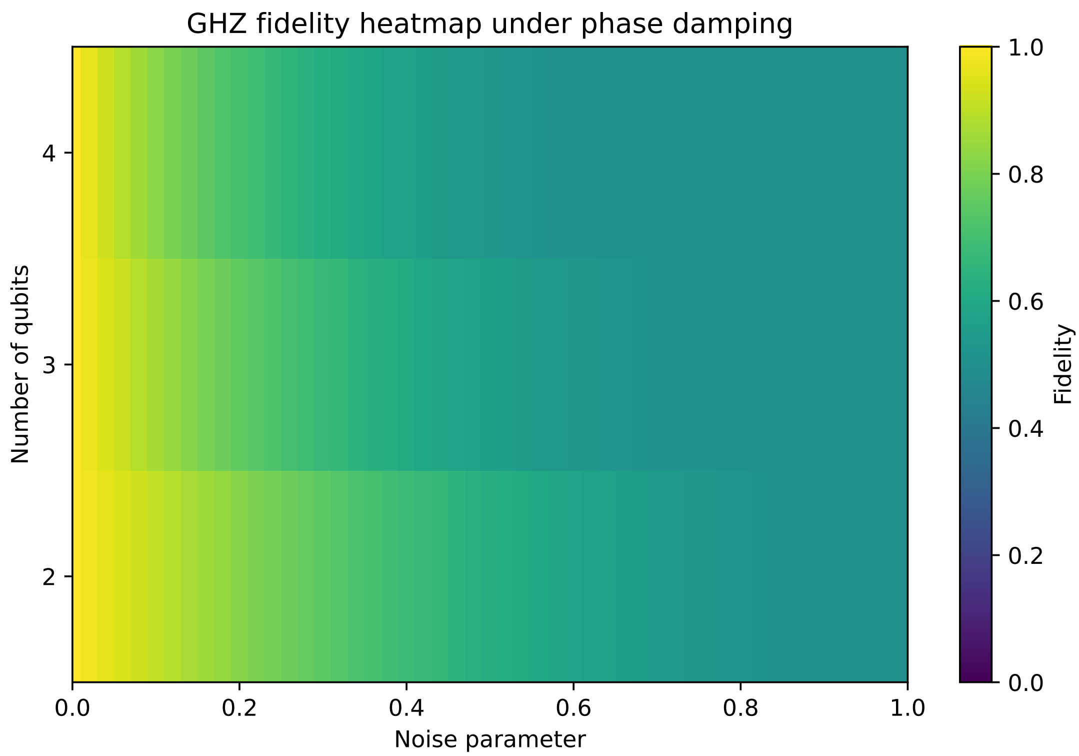

Figure 6: Fidelity heatmap for GHZ states under phase damping. The horizontal axis is the noise strength, the vertical axis is the qubit number, and the color scale represents fidelity.

A.3 Output inventory

The current implementation writes five CSV files and six SVG plots. Table 4 summarizes the exported artifacts.

B Selected Numerical Checkpoints

To support manual inspection of the saved data, Table 5 lists representative checkpoints drawn directly from the CSV outputs.

C Module Walkthrough

This appendix connects specific mathematical roles to specific source locations.

C.1 `core/states.py`

This module defines the basis-state constructor, the density-matrix conversion routine, a general tensor-product utility, and the three supported state families. The GHZ constructor enforces the range $2 \leq n \leq 4$, making the qubit limit an executable rule rather than a documentation remark.

C.2 `core/noise.py`

This module contains the core channel logic. It validates that γ and λ remain in $[0, 1]$, defines the Kraus operators of Eqs. (12) and (13), embeds single-qubit operators into n-qubit space, and stabilizes the resulting density matrix by restoring Hermiticity and renormalizing the trace.

Table 4: Output files generated by the bundled experiment suite.

Output type	Files

CSV data	`single_qubit_amplitude_sweep.csv`
	`bell_state_amplitude_sweep.csv`
	`bell_state_phase_sweep.csv`
	`ghz_phase_scaling.csv`
	`ghz_phase_fidelity_heatmap.csv`

SVG plots	`single_qubit_amplitude_fidelity.svg`
	`single_qubit_amplitude_purity.svg`
	`bell_state_fidelity_comparison.svg`
	`bell_state_purity_comparison.svg`
	`ghz_phase_scaling.svg`
	`ghz_phase_fidelity_heatmap.svg`

Table 5: Selected numerical checkpoints from the exported CSV files.

Study	Parameter value	Fidelity	Purity
Single-qubit amplitude damping	$\gamma = 0.00$	1.0000	1.0000
Single-qubit amplitude damping	$\gamma = 0.50$	0.85355	0.8750
Single-qubit amplitude damping	$\gamma = 1.00$	0.5000	1.0000
Bell amplitude damping	$\gamma = 0.50$	0.6250	0.5625
Bell phase damping	$\lambda = 0.50$	0.6250	0.53125
GHZ phase damping ($n = 2$)	$\lambda = 0.35$	0.71125	0.589253125
GHZ phase damping ($n = 3$)	$\lambda = 0.35$	0.6373125	0.5377094453125
GHZ phase damping ($n = 4$)	$\lambda = 0.35$	0.589253125	0.5159322406445312

C.3 `core/metrics.py`

This module checks the physicality conditions of Eqs. (3)–(5) numerically and computes the fidelity and purity values used throughout the experiment suite.

C.4 `core/evolve.py`

This module performs the actual state evolution under local noise. It chooses the channel based on the string label `"amplitude"` or `"phase"`, verifies the system dimension, and then applies the local channel to each qubit sequentially.

C.5 `experiments/sweep.py`

This module packages the evolution logic into reusable sweeps. It supports both parameter sweeps over channel strength and scaling sweeps over qubit number. It also allows both zero-argument and one-argument state constructors, which is why the same sweep code can handle both the fixed Bell-state function and the parameterized GHZ-state function.

C.6 `experiments/run_experiments.py` and `visualization/plots.py`

Together, these modules define the bundled studies, write the CSV outputs, generate the plots, and keep the output directory organized. The heatmap writer stores the matrix together with labeled axes, allowing direct reuse in later analysis.

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